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The impact of large-scale agricultural investments on welfare of local communities: a meta-analysis

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Large-scale agricultural investments (LSAIs) are promoted as pro-poor investments by governments in developing countries. However, their impact on local communities has remained contested due to concerns that these investments are associated with land concessions and displacements. A recent body of literature has documented mixed effects, leaving their broader impact inconclusive. This meta-analysis aimed to understand the overall effect of these investments on local communities. We computed the average effect size using partial correlation coefficients (pcc). The analysis is conducted on 39 primary studies obtained from Web of Science, Scopus and Google Scholar. To account for dependencies among effect sizes arising from multiple estimates within single studies, we used robust variance estimation (RVE) to estimate the average effect size. The results revealed a statistically positive, but marginal, average pcc of 0.043. This implies that LSAIs explain only less than one percent of the variation in welfare of local communities, suggesting that LSAIs have limited practical impact on improving living standards of local communities. The positive effect, though marginal, is achieved through the pathways of the role of LSAIs to improve income and food security. The analysis also indicated that the impact on local welfare varies based on the different LSAI models. Out-grower and contract farming models of LSAI appear to yield more positive outcomes to the local community compared to plantation-based LSAI models. Given the positive but marginal overall effect, policy makers should exercise caution when considering further LSAIs in developing countries to maximize potential benefits by mitigating its negative impacts. Future research could explore cross-country heterogeneity, long-term impacts, national-level effects, and non-operational LSAIs to better inform policy.

Introduction

Agricultural production in many developing countries remains low amid substantial growth potential. Exploiting this potential has been promoted as a strategy to reduce poverty in rural areas where most of the world's poorest people live (FAO (2012); World Bank (2008)). Achieving this growth potential requires significant investments in fertilizers, improved seeds, and infrastructure such as roads and markets. To address these, governments in

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developing countries have encouraged private sector participation including foreign investors through large-scale land investment programs, offering incentive packages like tax breaks and loans. This initiative aligns with a surge in global commodity and food prices, driving higher demand for land for food production (Borras Jr & Franco, 2012; Deininger, 2011; Edelman et al., 2013). Consequently, in the last couple of decades, large scale agricultural investments (LSAIs) have increased dramatically. According to the database of the Land Matrix [1], since the early 2000, more than 65 million hectares of agricultural land-an area larger than France (64.5 million ha) and nearly twice the size of Germany (35.7 million ha) - have been transferred from local communities to private and public investors through large-scale land acquisitions (Land Matrix (2023)).

While the overall goal of large-scale agricultural programs is often to stimulate local development and reduce poverty, their impact on the welfare of local communities remains contested, with evidence yielding both positive and negative outcomes. On the positive side, LSAIs can create employment opportunities in rural areas where formal job opportunities are limited. The wages earned from such jobs contribute to household income, improve living standards (Anti, 2021; Baumgartner et al., 2015; Bottazzi et al., 2018) and enhance household food security (Fitawek & Hendriks, 2021). Additionally, LSAIs often generate spillover effects, such as access to improved agricultural inputs and technologies, which enhance local productivity, food availability, and overall food security (Ali et al., 2019; Herrmann, 2017; Sullivan et al., 2022). On the other hand, studies report negative consequences associated with LSAIs. These investments compete with local communities for land, leading to dispossession and displacements (Messerli et al., 2014; Lay et al., 2021) which in turn diminish agricultural productivity, reduce income and exacerbate food insecurity (Aisbett & Barbanente, 2016; Guyalo et al., 2022b; Shete & Rutten, 2015). Furthermore, LSAIs are often linked to environmental degradation, including loss of forest resources, and soil degradation, which further jeopardize local income sources (Jiao et al., 2015; Schoneveld, 2014). However, these studies offer a fragmented view of LSAIs' impact, often relying on case studies of individual investment projects. This fragmented evidence complicates efforts to draw generalizable conclusions about the broader effects of LSAIs. Synthesized evidence that provides a comprehensive perspective on the impact of LSAIs on local community well-being could address this gap.

However, a review of the literature reveals a dearth of meta-analytic studies that provide comprehensive evidence on the impact of LSAIs on the welfare of local communities. Nevertheless, a few qualitative systematic reviews with specific thematic or geographic focus exist (Borras et al., 2022; Cochrane & Legault, 2020; Hufe & Heuermann, 2017; Jung, 2018; Rasva & Jürgenson, 2022; Yang & He, 2021). Borras et al. (2022), in a review of 28 studies on the failed large-scale land investments, underscored that the causes and consequences of the global land rush remain only partially understood. Cochrane & Legault (2020), after reviewing 71 studies on agricultural land investments in Ethiopia, noted gaps in research on the gendered impacts of LSAIs and the role of diaspora and domestic investors. Jung (2018) and Yang & He (2021), assessed research designs in primary research; Jung found that adverse effects on livelihoods are more often reported in qualitative studies, as these studies focus on investigating the negative effect of land dispossessions. Yang & He highlighted the lack of focus on regional-level impacts compared to community-level assessments. The other two reviews emphasize the welfare and livelihoods impact of LSAIs. Hufe & Heuermann (2017) reviewing 60 case studies from 22 countries from continental Africa, and Rasva & Jürgenson (2022) reviewing 40 studies from Europe, both studies highlighted the predominantly negative effect of land investments on the welfare of host communities. Detailed presentation of the systematic reviews is provided in the supplementary material.

In this study, we conducted a meta-analysis to provide comprehensive evidence on the overall impact of LSAIs on welfare of local communities, using 195 estimates collected from 39 primary studies. The findings reveal a statistically positive but practically negligible effect of LSAIs on welfare of local communities. To explore heterogeneity in the primary studies, meta-regression and subgroup meta-analysis were employed. The results provide valuable insights for policy makers, to enhance the implementation of large-scale land investments and maximize benefits for the local communities. The rest of the paper is organized as follows: Section two outlines the conceptual framework; section three details the data and methods; section four presents the results and discussions and the last section provides conclusions.

LSAIs and welfare of local communities

Theoretical framework. Large-scale agricultural investments (LSAIs) could significantly influence welfare of local communities in both positive and negative ways. These impacts manifest through economic, social, and environmental channels. During the initial proliferation of LSAIs, proponents argue that LSAIs boost welfare by creating jobs, enhancing productivity, developing infrastructure, and generating tax revenues (Deininger & Byerlee, 2011). Critics, however, highlight risks such as resource restrictions, displacement, and environmental harm (Cotula, 2009; Deininger, 2011). One of the earliest theoretical explanations of the impact pathways of LSAIs on the welfare of local communities is illustrated by Dessy et al. (2012). They developed an occupational choice model based on the assumption that smallholders affected by LSAIs can either remain in the farming community and share the remaining land or switch to wage employment on the investment farm, opting for the choice that offers the higher payoff. According to their model, LSAIs improve the welfare of local communities under the conditions that: first, the state negotiates deals that are beneficial to local people; second, these deals create sufficient job opportunities; and third, wage employment earnings provide displaced farmers better livelihoods than their previous land-based occupations.

Building on this, Kleemann & Thiele, (2015) extended Dessy et al.'s occupational choice model to further conceptualize the pathways through which LSAIs affect local welfare. Their theoretical framework indicated that the level of compensation, employment effects, and spillover effects are the main pathways through which LSAIs affect local communities. They argue that the extent of labor demand from LSAI farms and the availability of spillover effects—such as technology transfer from LSAI farms to local farmers—are the primary determinants of whether LSAIs lead to welfare improvements for local communities. Both of the theories emphasize the importance of adequate compensation, job creation, and knowledge and technology transfer to improve living conditions of the affected communities.

Another widely used theoretical framework in the empirical literature that indicate how LSAIs affect welfare outcomes of local communities is the Sustainable Livelihood Approach (SLA). Studies employ SLA (see for e.g., Alhassan et al., 2018; Bosch & Zeller, 2019; Talleh Nkobou et al., 2021) to examine how policies and programs like LSAIs impact welfare outcomes such as income, food security, asset accumulation, and resource sustainability. The SLA posits that livelihood outcomes result from the interaction of various components, including livelihood capitals (human, social, physical, natural, and financial), strategies (farm and non-farm activities), and the institutional and policy context (Serrat, 2017). Figure 1 explains the potential channels through which LSAIs could affect welfare of local communities.

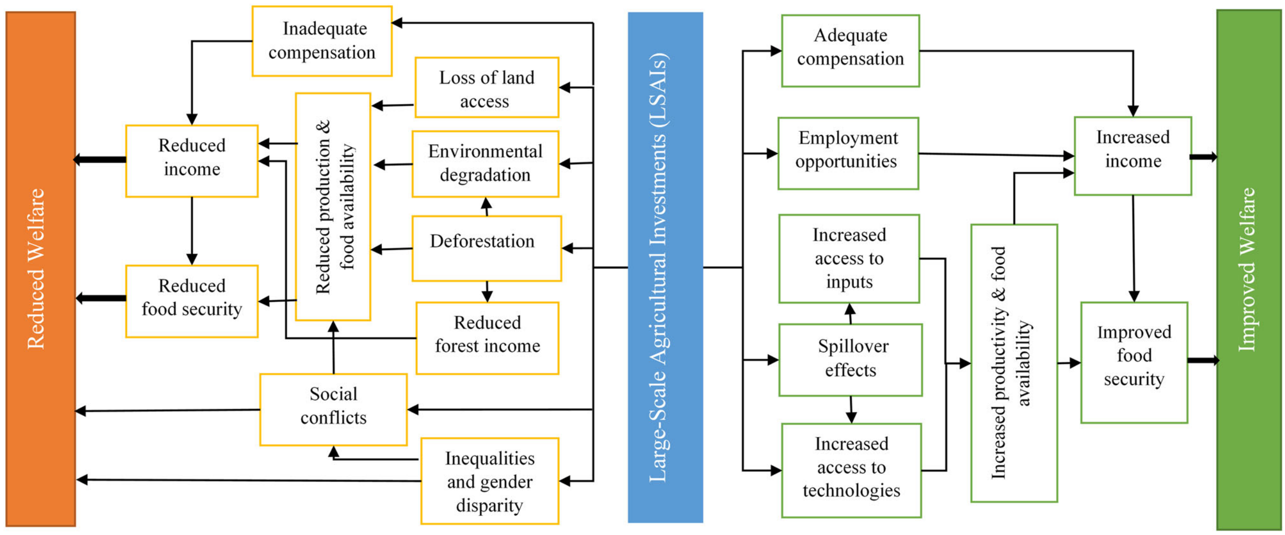

Fig. 1 Theoretical framework of the study. It shows the impact pathways through which large-scale agricultural investments affect welfare of local communities.

Empirical evidence. Over two decades of research on Large-Scale Agricultural Investments (LSAIs) indicated that LSAIs pose mixed impacts on local communities. Consistent with the theoretical frameworks, these studies show that LSAIs influence welfare through multiple pathways. The positive effects are demonstrated by the contribution of LSAIs to; (i) increasing income and employment opportunities: LSAIs create rural jobs including shifting households from traditional farming to wage-dependent livelihoods, thereby improving income and living standards (Anti, 2021; Baumgartner et al., 2015; Bottazzi et al., 2018). These employment opportunities enhance household food security (Fitawek & Hendriks, 2021). (ii) Improving agricultural productivity and food availability: LSAIs enhance food availability by increasing agricultural output, reduce prices, and boost local productivity through spillover effects in the form of access to improved inputs and technologies (Ali et al., 2019; Herrmann, 2017; Makunike & Kirsten, 2018; Sullivan et al., 2022). (iii) Poverty reduction: LSAIs also contribute to poverty reduction by developing infrastructure such as roads, irrigation systems, and storage facilities. In Laos, for instance, LSAIs reduced poverty of rural communities affected by LSAIs (Nanhthavong et al., 2020).

On the other hand, the negative effects of LSAI on the local community manifest through the channels (i) loss of land and resource rights: LSAIs often require significant amounts of land, leading to the dispossession and displacement of local communities. These investments tend to concentrate on accessible, productive, and densely populated areas than idle and remote areas (Messerli et al., 2014), competing over land with small holders (Lay et al., 2021). This has led to reduced agricultural productivity, income and food insecurity (Aisbett & Barbanente, 2016; Guyalo et al., 2022b; Shete & Rutten, 2015). (ii) Environmental degradation and resource loss: Large-scale land clearing for agriculture frequently causes deforestation, loss of forest resources, and soil degradation, depriving communities of essential ecosystem services (Jiao et al., 2015; Schoneveld, 2014). (iii) Inadequate compensation and social tensions: Compensation provided to displaced communities is often insufficient to offset the loss of land and livelihoods (Cotula, 2009; Deininger & Xia, 2016) and causes displacements (Lay et al., 2021). This has led to tensions between investors and local populations, eroding social cohesion and increasing poverty (Borras Jr & Franco, 2012). Recent evidence on this regard also showed that many LSAIs provoke conflicts, as they disregard community interests, sparking peaceful protests that often escalate into coercion and violence by armed actors linked to governments and investors (Juan et al., 2022). (iv) Inequalities and gender disparities: opportunities created by LSAIs are not always equally accessible to women. For instance, employment opportunities from LSAIs tend to favor men, deepening gender inequality (Bottazzi et al., 2018; Hajjar et al., 2020). Despite the potential benefits and drawbacks of LSAIs, their overall impact, which should guide policy decisions, remains unclear. In this meta-analysis, leveraging both the theoretical frameworks and empirical evidence, we document the comprehensive impact of such investments on local communities.

Method of analysis

Literature search and screening strategy. The procedures of this study follow the guidelines established by the Meta-Analysis of Economics Research Network (MAER-Net) (Havránek et al., 2020)[2].The search strategy is designed to address the research question: *What is the impact of LSAIs on welfare of local communities*? This question guides the development of search strings constructed based on the population, intervention, outcomes, and study designs (PIOS) framework. The Population is represented by countries hosting LSAIs. The intervention is LSAIs; the outcome is one of the welfare indicators such as income, consumption, food security, asset, and others, and the study design is quantitative studies. However, the search query doesn't directly include population and study design criteria. Rather, we used them at the screening stage. Consequently, the search string contains two keyword sets: LSAIs and welfare indicators. Terms within each set are connected by "OR," while the two sets are combined using "AND." The full list of keywords used is available in Appendix (Table A1). The search string was applied across digital bibliographic databases, including the Web of Science Core Collection, Scopus, and Google Scholar. Additionally, a manual snowballing technique was employed to identify relevant studies through systematic reviews and selected primary studies. The search process concluded on December 12, 2022.

Our search focuses on identifying quantitative studies that estimate a model on the welfare impacts of LSAIs as follows:

$$W_{ij} = \alpha + \beta(LSAI_j) + \varepsilon_{ij} \tag{1}$$

where i represents the estimate obtained from study j; W represents welfare outcomes (example, income, food security,

Table 1 Included studies in the meta-analysis and number of estimates in each study.

Study Name	Number of estimates	Study Name	Number of estimates	Study Name	Number of estimates
Aisbett & Barbanente, (2016)	16	Guyalo et al. (2022a)	8	Osabuohien et al. (2019)	6
Akyoo et al. (2018)	2	Guyalo et al. (2022b)	5	Osabuohien et al. (2020)	1
Alamirew et al. (2015)	2	He et al. (2021)		Quansah et al. (2020)	1
Alemu & Tolossa, (2022)	3	Herrmann (2017)	12	Schüpbach (2015)	12
Alhassan et al. (2018)	4	Herrmann & Grote (2015)		Shete & Rutten (2015)	4
Anti, (2021)	15	Hofman et al. (2019)	20	Sullivan et al. (2022)	4
Antonio (2015)	4	Jeckoniah et al. (2020)	1	Talleh Nkobou et al. (2021)	1
Bekele et al., (2021)	5	Jiao et al. (2015)	6	Tan et al. (2021)	5
Bekele et al., (2021)	1	Karakara et al., (2022)	1	Thornhill (2016)	2
Bosch & Zeller (2019)	1	Kebede et al. (2021)	3	Tuyen et al. (2014)	4
Bottazzi et al. (2018)	3	Makunike & Kirsten (2018)	2	Wayessa (2020)	2
Fitawek & Hendriks (2021)	12	Nanhthavong et al. (2020)	1	Wang et al. (2019)	6
Guyalo et al. (2021)	5	Nanhthavong et al. (2021)	4	Wendimu (2015)	4

expenditure, asset, or poverty). $LSAI_j$ represents large-scale agricultural investment exposure in study j usually a binary variable. β captures the impact of LSAI exposure on welfare and ε_{ij} is the error term.

Primary studies are included in the meta-analysis dataset if they meet three criteria. First, the paper should be written in English language. Second, the land for LSAI considered by the primary studies should be transferred after 2000, coinciding with the surge in large-scale land acquisitions. Third, it should be conducted in one of the countries in the global south where LSAIs proliferate. Fourth, it should quantitatively estimate the effect of LSAIs on welfare of local communities. The Prisma flow diagram that portrays the identification and screening procedures is presented in the appendix (Fig. A1). Using the search query and inclusion criteria presented above, we selected 39 studies to be included in the meta-analysis data set. Each study contributed from one to 20 estimates and on average 5 estimates to the database (Table 1).

Estimates are extracted and coded from the 39 primary studies. Since studies report multiple estimates, we followed an "all-set" approach to extract and code 195 observations[3]. The approach takes all the estimates reported by the author. By doing so, this approach has the following advantages: increase the number of observations for meta-analysis and avoid bias arising from the selection of the best estimate (Stanley & Doucouliagos, 2012). The potential interdependencies between data points could pose an estimation challenge. However, it can be handled by employing appropriate statistical methods such as robust variance estimation (RVE) method.

We collected data on the point estimate of the coefficient β ; inferential statistics of the point estimate such as t-statistic, standard error, sample size, degrees of freedom, and sign of the estimate; research design and estimation technique, and publication characteristics. Data extraction is made on a predetermined excel form by the lead author. To minimize potential mistakes during the data extraction process, one of the coauthors performed checks on random portions of the data. Besides, existence of outliers and leverage points is also checked. Outliers are extreme and implausible values of the effect size, whereas leverage points are extreme values of the precession, which would cause the bias in the true effect size (Stanley & Doucouliagos, 2012). To ensure that outliers do not bias the true effect, we winsorize the collected estimates and standard errors at the 5% level following Zigraiova et al., (2021). As a robustness check we conducted the meta-analysis without winsorizing and by excluding the influential observation. The results for this exercise are presented in the supplementary material.

Description of the data is presented in Table 2. These variables are presented using six groups. The first is data characteristics: Closer to 70% of the estimates in our meta-analysis are derived from cross-sectional data. Additionally, the age of the farm, calculated as the difference between data collection year and land transfer year may impact the estimated effects. Evaluations conducted shortly after the establishment of a farm may show a negative impact, as the benefits to the community often emerge gradually over time. To address this, we collected information on the year of land transfer and the year of data collection. The second is estimation characteristics. The primary studies used different estimation methods to evaluate the impact of LSAIs on local welfare. Most of the studies employed impact evaluation techniques, from which propensity score matching is the most frequently utilized method, accounting for 52.8% of the studies. 26.7% of the studies used difference-in-differences (DID) method. The remaining studies used methods such as Ordinary Least Squares (OLS) regression, ANOVA, and correlation analyses are also applied. The third is type of outcomes: Primary studies employed a range of indicators of welfare, with food security, income, and asset accumulation being the most frequently used indicators. The fourth is host countries: the primary studies included in the synthesis are conducted in 15 different countries. Among which Ethiopia and Tanzania are the most studied countries, representing 30.8% and 14.4% of the included estimates, respectively.

The fifth is Publication characteristics: The peer review process can influence the magnitude of effect sizes reported in primary studies. To account for this, variables such as publication status, and publication year were included in the analysis. Closer to 72% of the estimates in the meta-analysis originated from peer-reviewed published studies. The last is LSAI Scheme: The impact of LSAIs on local community welfare could differ depending on the production arrangements adopted. Large-scale land acquisitions and more integrated production models, such as out-grower schemes and contract farming platforms, exhibit varying effects. In the meta-analysis database, about 22% of the estimates are derived from studies evaluating the impacts of contract farming and out-grower production.

Table 2 Description and summary of selected coded variables.

Variable	Description	N	mean	sd
pcc	Partial correlation coefficient	195	0.0548	0.207
Data characteristics				
sepcc	Standard error of pcc	195	0.0470	0.0248
Data type	=1 if cross sectional data is used	195	0.697	0.461
Age of the farm	Difference between data collection and land transfer years	195	5.841	4.478
Estimation				
Propensity score matching	=1 if propensity score matching is used	195	0.528	0.500
Difference-in-difference	=1 if difference in difference is used	195	0.267	0.443
Ordinary least squares	=1 if ordinary least square is used	195	0.103	0.304
Other methods	=1 if ANOVA, Correlations, and IV are used (reference category of estimation method)	195	0.103	0.304
Outcomes				
Asset	=1 if the outcome variable is asset	195	0.195	0.397
Employment	=1 if the outcome variable is employment	195	0.0718	0.259
Food security	=1 if the outcome variable is food security	195	0.313	0.465
Income	=1 if the outcome variable is income	195	0.246	0.432
Livelihoods	=1 if the outcome variable is livelihoods	195	0.0718	0.259
Other outcomes	=1 if the other outcome variable is used (reference category of outcomes)	195	0.103	0.304
Countries				
Cambodia	=1 if the LSAI host country is Cambodia	195	0.108	0.311
China	=1 if the LSAI host country is China	195	0.0615	0.241
Ethiopia	=1 if the LSAI host country is Ethiopia	195	0.308	0.463
Sierra Leone	=1 if the LSAI host country is Sierra Leone	195	0.118	0.323
Tanzania	=1 if the LSAI host country is Tanzania	195	0.144	0.352
Zambia	=1 if the LSAI host country is Zambia	195	0.0718	0.259
Other countries	=1 if the LSAI host countries are others (reference category of countries)	195	0.190\	0.393
Author affiliation	=1 if the lead author is affiliated to African institutions	195	0.313	0.465
Publication				
Publication status	=1 if the study is published	195	0.723	0.449
Relative publication year	Pub. year minus the earliest pub. year (2014)	195	4.569	2.648
Citation count	Number of citations in google scholar	195	26.22	35.33
LSAI scheme				
Out grower scheme	=1 if out grower or contract farming schemes are used	195	0.221	0.416

N represents the number of observations, and sd denotes the standard deviation. Outcomes in other outcomes variable include inequality, health, empowerment and resilience. Countries with small number of estimates reported are categorized under other countries. These countries include Mozambique, Nigeria, Ghana, Vietnam, Kenya, Laos, Myanmar, Malawi, Madagascar.

Estimation strategy

Meta-analytic approach. The average estimated value of β in Eq. (1) represents the impact of LSAIs on local communities' welfare. However, comparing these estimates across primary studies is challenging because different studies use different welfare measures. To address this issue, we standardized the estimated value of β using the partial correlation coefficient (PCC). It is a widely used metric in economic meta-analysis (see for e.g., Cipollina et al., 2018; Doucouliagos, 2005; Doucouliagos & Ulubasoglu, 2006; Efendic et al., 2011; Valickova et al., 2015). The PCC indicates the degree and direction of association between two variables providing a unitless measure that facilitates comparison across studies (Borenstein et al., 2009; Stanley & Doucouliagos, 2012). It offers two key advantages: (i) it is unitless comparison: PCC allows direct comparison of studies regardless of the welfare indicators or measurement units used, and (ii) simplicity and interpretability: the PCC is straightforward to define and interpret, making it an effective standard metric. However, the PCC, as an index of effect size, is interpreted differently than a simple correlation coefficient. Specifically, the squared PCC (PCC^2) represents the proportion of variance in the welfare indicators explained by LSAIs (Card, 2012). For instance, a PCC of 0.5 implies that, *ceteris paribus*, 25% (i.e., 0.5^2) of the variance in welfare outcomes is explained by LSAIs.

Most primary studies rarely report PCC. Hence, partial correlation coefficients are computed from reported regression statistics such as t-statistics and degrees of freedom as follows:

$$r_{ij} = \frac{t_{ij}}{\sqrt{t_{ij}^2 + df_{ij}}} \tag{2}$$

where r_{ij} represents the PCC of the i^{th} estimate from the j^{th} study; t_{ij} denotes t-statistic of the i^{th} estimate from the j^{th} study and df_{ij} is the degrees of freedom of the t-statistic (Stanley & Doucouliagos, 2012). The standard error of the partial correlation coefficient is calculated as $SEr_{ij} = \sqrt{(1 - r_{ij}^2)/df_{ij}}$ where SEr_{ij} is standard error of the partial correlation coefficient, r_{ij} and t_{ij} are as defined earlier.

In meta-analyses, before the PCC is used to compute mean effects, it undergoes Fisher's r-to-z transformation (Z_r). This is because the PCC does not follow normal distributions, especially when its value gets closer to $+1$ and -1. The transformation ensures symmetry, which is crucial for accurately computing mean effects. The formula for Z_r is: $Z_r = 0.5ln(1 + r_{ij}/1 - r_{ij})$ where Z_r is the Fisher's transformed PCC and r_{ij} is as defined in Eq. (2). While Z_r is useful for statistical analysis, it lacks interpretability since its values are not bounded to $+1$ and -1. To enhance interpretability, Z_r is transformed back to PCC for reporting purposes (Card, 2012).

The simple average PCC computed using Eq. (2) assigns equal weight to individual estimates and does not account for precision,

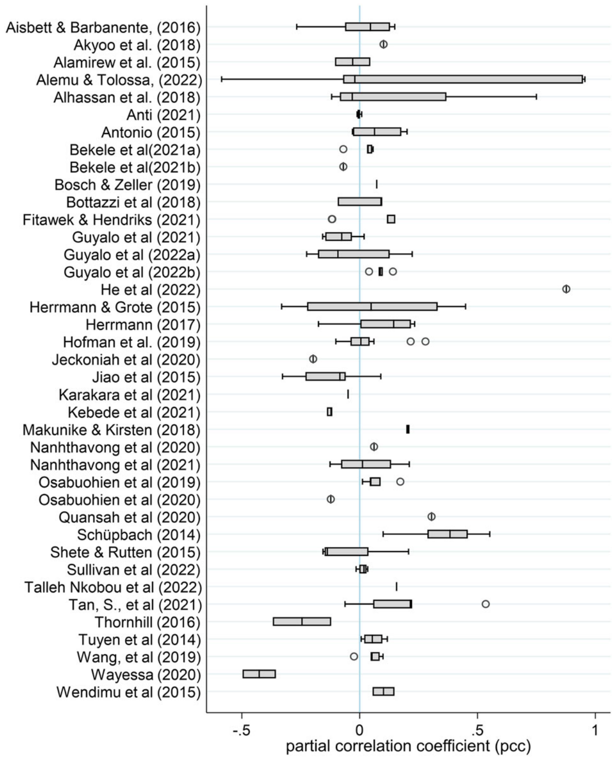

Fig. 2 The figure shows box plot of the partial correlation coefficients computed from estimates of individual studies. The box plots indicate that estimates vary both within and across studies. The length of each box represents the interquartile range (P25-P75), and the vertical line inside the box indicates the median value. The whiskers represent the highest and lowest data points within 1.5 times the range between the upper and lower quartiles. The vertical line denotes zero partial correlation coefficient. Circles represent outliers.

typically measured by standard errors. To address this limitation, a precision-weighted mean PCC is calculated using random-effects model (Borenstein et al., 2009; Card, 2012). Let r_{ij} denote effect size estimate i from study j, with corresponding standard error s_j. Assuming that r_{ij} is unbiased estimate of the effect size parameter θ_{ij}; X_{ij} denote a row vector of covariates (including an intercept term) and α denote a vector of regression coefficients, the effect size is estimated using the meta-regression model;

$$r_{ij} = X_{ij}\alpha + \mu_j + \nu_{ij} + e_{ij} \tag{3}$$

where r_{ij} is as defined earlier, $var(\mu_j) = \tau^2$ (between study variance); $var(\nu_{ij}) = \omega^2$ (within study variance) and $var(e_{ij}) = s_j^2$. The average PCC in this case can be obtained from each studies' estimate weighted by the inverse variance of the sum of within and between study variance:

$$w_{ij} = \frac{1}{\omega_{ij}^2 + \tau_j^2} \tag{4}$$

where; w_{ij} represents the random effect weight, tau-squared (τ^2) is the between study variance and ω_{ij}^2 captures the within study variance (Pustejovsky & Tipton, 2022).

Studies included in the synthesis reported different welfare outcomes on the same samples, resulting in 1 to 20 effect sizes per study (Table 1). This results in within-study dependencies and correlated errors which poses a challenge for conventional meta-analytic approaches (Hedges et al., 2010; Tipton, 2015). To address these dependencies, a robust variance estimation (RVE) meta-analytic approach is employed to estimate Eq. (3), which allows for multiple indicators from a single study (Hedges et al., 2010; Tanner-Smith et al., 2016). The advantage of RVE is that it produces consistent point estimates, standard errors, confidence intervals, and significance tests without requiring explicit modeling of the dependency structure (Tipton, 2015). Given the limited knowledge regarding whether the data has correlated or hierarchical structure among effect sizes, Correlated and Hierarchical Effects (CHE) working model for RVE is selected. The CHE working model combines the desirable features of both correlated and hierarchical effect models and produces robust results (Pustejovsky & Tipton, 2022). RVE is broadly applicable in social science systematic reviews (Pustejovsky & Tipton, 2022). As a robustness check, results from the correlated effects (CE) and hierarchical effect (HE) working models are reported in the supplementary material.

The random effects model is estimated using metafor package using the "rma" function (Viechtbauer, 2010) and clubSandwich (Pustejovsky, 2020) R packages to estimate the mean effect size, with variance components derived via restricted maximum likelihood (REML). However, since the impute covariance matrix is deprecated in clubSandwich 0.5.11, we used the metafor::vcalc instead.

Publication bias test. While publication bias is a concern in economics research (Ioannidis et al., 2017; Stanley, 2008; Stanley & Doucouliagos, 2012, 2014), our primary objective is not to test the extent of publication bias in the LSAIs literature. Instead, this study aims to contribute to the ongoing debate by estimating the overall effect of LSAIs on local communities. However, to align with standard meta-analysis practices and observe the publication-bias-corrected PCC, we conducted publication bias tests using both linear and non-linear methods.

First, we conducted a publication bias test using a funnel plot (Egger et al., 1997; Stanley, 2005). Second, we quantify the extent of publication bias using linear and non-linear publication bias test techniques. We applied the linear numerical test using the funnel plot asymmetry test and the precision-effect test (FAT-PET) (Stanley, 2005). The test, which is conducted based on Eq. (5), assumes that, in the absence of publication bias, the PCC will be independent of its standard error. Conversely, if publication bias is present, a correlation between the PCC and its standard error will be observed. This method has been widely used in the meta-analysis literature to detect and correct for publication bias (see, for e.g., Havranek & Irsova, 2011; Havranek & Kokes, 2015; Stanley, 2005; Valickova et al., 2015; Zigraiova et al., 2021) .

$$r_{ij} = \beta_1 + \beta_0(SE_{ij}) + \varepsilon_{ij} \tag{5}$$

where; r_{ij} represents the partial correlation coefficient of the i^{th} outcome variable in the j^{th} study. The coefficient β_0 indicates publication bias; if β_0 is statistically significant, it provides valid evidence for the presence of publication bias. On the other hand, β_1 measures the true effect size corrected for publication bias (Stanley, 2005). However, the specification in Eq. (5) is heteroscedastic, as right-hand-side variable (SE_{ij}) captures the variance of the left-hand side variable (r_{ij}). To address this heteroscedasticity, we divide Eq. (5) by the standard error following the standard practice in met-analysis literature (e.g., Havranek & Irsova, 2011; Havranek & Kokes, 2015; Stanley, 2005; Valickova et al., 2015; Zigraiova et al., 2021), resulting in the following transformation:

$$t_{ij} = \beta_0 + \beta_1 * \frac{1}{(SE_{ij})} + \omega_j + \varphi_{ij} \tag{6}$$

where t_{ij} represents the t-statistic of the i^{th} outcome variable in

Table 3 Meta-analysis results from the robust variance estimation (RVE) of the impact of LSAIs on welfare of local communities.

Variables	k	PCC	SE	t-value	dfs	I^2(%)	95% CI
Panel A: Overall effect of LSAIs on welfare							
Overall effect	195	0.0431**	0.0198	2.18	34.4	66.7%	[0.0043, 0.0819]
Overall effect (with covariates)	-	0.0212	0.022	0.965			[−0.0219, 0.064]
Panel B: The effect of LSAIs on specific welfare indicators							
Income	48	0.0508*	0.0279	1.82	16.2	46.23%	[−0.0039, 0.1054]
Food Security	61	0.0262*	0.0141	1.87	10.3	29.56%	[−0.0091, 0.0616]
Employment	14	0.0922	0.0727	1.27	4.91	53.28%	[−0.0517, 0.2360]
Asset	38	−0.0195	0.0685	−0.285	7.92	74.78%	[−0.1539, 0.1149]
Panel C: The effect of LSAI on welfare based on LSAI schemes							
Plantation	152	0.0205	0.0139	1.48	14.3	61.7%	[−0.0070, 0.0480]
Out grower	43	0.0872	0.0561	1.56	7.72	65%	[−0.0231, 0.1974]

Notes. All models were run separately and estimated using robust variance estimation meta-analysis. Overall effect represents the RVE estimate with an intercept only model. Overall effect (with covariates) represents the overall effect size after accounting all the covariates (see section 4.6 for details). Models with degrees of freedom <4 are not trustworthy.
k number of estimates, *PCC* partial correlation coefficient, *SE* robust standard error, *dfs* degrees of freedom, *CI* confidence interval.
**$p < 0.05$, and *$p < 0.10$.

the j^{th} study; ω_j denotes study-level random effects and φ_{ij} captures estimate level disturbances.

We control the potential within-study dependencies that arise due to the inclusion of multiple estimates per study by using a mixed-effects model to estimate Eq. (6). This approach balances the weights assigned to studies, particularly when between-study heterogeneity is significant (Rabe-Hesketh & Skrondal, 2008). Besides, standard errors are clustered at the study level to address within-study dependencies. However, clustering can perform well when clusters are uneven (i.e., when the number of estimates per study varies significantly) and when the total number of studies is fewer than 40 (Irsova et al., 2023). Under such conditions, bootstrapped standard errors are more robust. Consequently, we report bootstrapped standard errors rather than clustered ones. To account for unobserved study-level heterogeneity potentially correlated with the publication bias term, we also estimate Eq. (6) using fixed effects by including study-level dummies.

To further test for publication bias, we employed the precision-effect estimate with standard error (PEESE), a non-linear methods to test publication bias (Stanley & Doucouliagos, 2012, 2014), specified as:

$$t_{ij} = \beta_0 * (SE_{ij}) + \beta_1 * \frac{1}{(SE_{ij})} + \omega_j + \varphi_{ij} \tag{7}$$

where β_0 denotes publication bias, and β_1 the mean PCC corrected for publication bias.

Meta-analysis results and discussions

The overall impact of LSAIs on welfare. Figure 2 presents Partial Correlation Coefficients (PCCs) across individual studies, depicted as box plots. It indicates significant within-study and between study heterogeneity. Each box plot shows the range of PCC values reported within a single study: narrower boxes indicate relatively consistent results, while wider boxes suggest greater variability. The differences in the range of box plots across studies indicate substantial between-study heterogeneity in the median PCC values. The observed heterogeneity could suggest employing random-effects model in the meta-analysis.

The meta-analysis results estimated from Eq. (3) that show the overall effect of LSAIs on local communities' welfare are presented in Table 3. The table reports results obtained from running separate robust variance estimation (RVE) on the Correlated and Hierarchical Effects (CHE) working models based on Eq. (3). Panel A of Table 3 presents the intercept only model that indicates the overall effect of LSIAs on welfare of local communities. The result revealed a statistically significant, but small, overall effect size (PCC = 0.0431, $p = 0.029$), with a 95% confidence interval of [0.004, 0.082]. The result indicated a within-study variance ($\sigma^2 = 0.016$) corresponding to substantial within study heterogeneity ($I^2 = 66.7\%$) and between-study variance of ($\tau^2 = 0.008$) reflecting low between study heterogeneity ($I^2 = 31.6\%$). This signifies that a larger proportion of the heterogeneity was attributed to within-study variability compared to between-study variability. The high degree of heterogeneity implies that caution should be exercised while interpreting the overall effect size. Moderator and sub-group analyses may help to explain additional variance in the effect size. The overall effect (with covariates) represents the overall effect size obtained by including different covariates. Results indicated that including covariates (see Section "Meta-regression results" for details) reduced the effect size (PCC = 0.021, 95% CI [−0.022, 0.064]), though it remained non-significant. This implies that observed moderators account for part of the initial heterogeneity.

The PCCs reported in Table 3 are classified as very small. According to Doucouliagos (2011), the strength of effect sizes measured in partial correlation coefficients is categorized as large (greater than 0.33), medium (between 0.07 and 0.33), and small (less than 0.07). While statistically significant, the PCCs of 0.0212 to 0.0431 falls within the small category, suggesting a minimal association of LSAIs with local communities' welfare. Squaring these PCC values indicates that LSAIs explain only a very small portion (0.045% to 0.18%) of the changes in welfare which is too small to translate in to improvements in living standards of the community.

Policymakers promote LSAIs as drivers of economic and social transformation due to their potential in creating jobs, enhancing productivity through spillover effects, developing infrastructure, and generating tax revenues (Deininger & Byerlee, 2011). As outlined in the theoretical framework (Fig. 1), LSAIs could positively contribute to the local community by creating employment opportunities and increasing income (Anti, 2021; Baumgartner et al., 2015; Bottazzi et al., 2018) improving agricultural productivity and food availability (Ali et al., 2019; Herrmann, 2017; Makunike & Kirsten, 2018; Sullivan et al., 2022). However, the observed minimal effect sizes documented in this meta-analysis contradict these expectations, suggesting that the actual benefits are substantially lower than anticipated. Consequently, a cautious approach is required when evaluating the developmental potential of LSAIs. Furthermore, these small PCCs support the critiques that the potential benefits of LSAIs are overstated (Lay et al., 2021) reinforcing existing concerns that these investments require extensive land acquisition, leading to resource restrictions, displacement, and

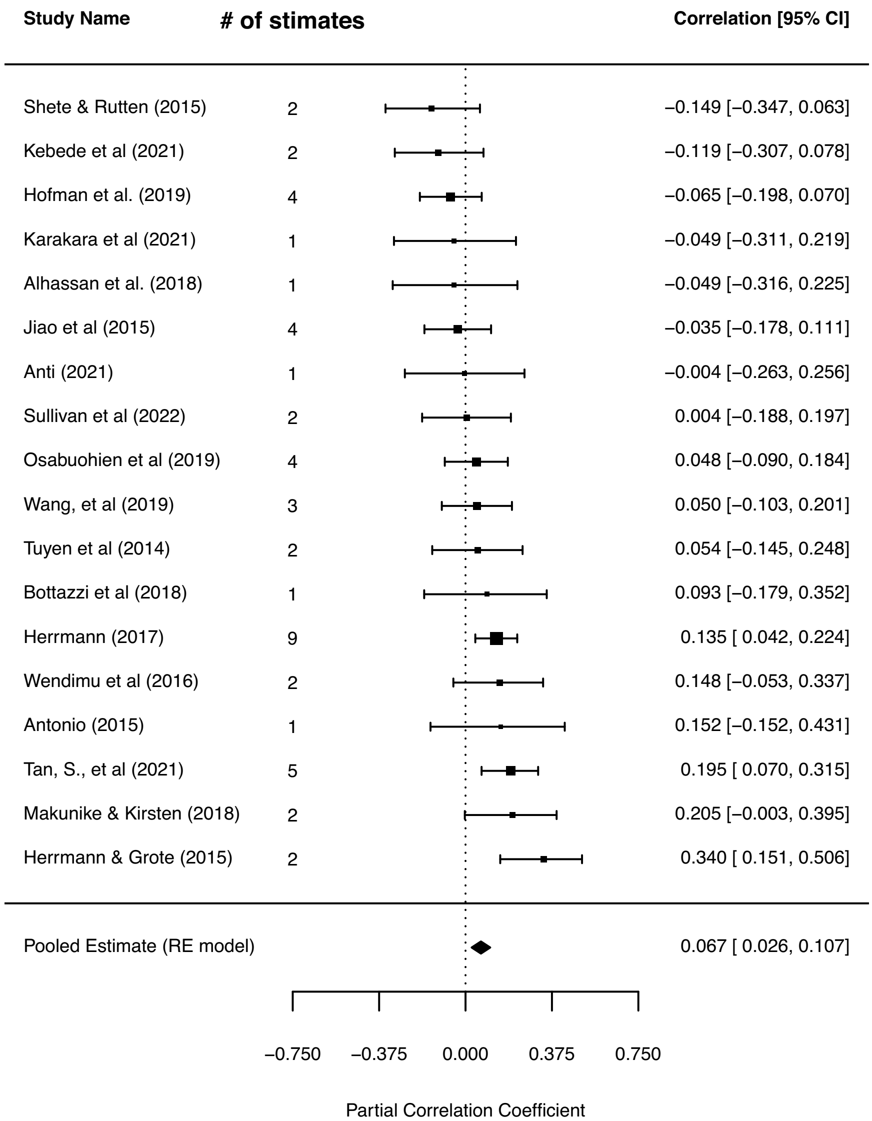

Fig. 3 Forest plot for random effects meta-analysis. It indicates the average effect of LSAIs on income of local communities.

environmental degradation (Cotula, 2009; Deininger, 2011; Deininger & Xia, 2016).

Despite this, the positive PCC documented in this study suggests that LSAIs could achieve their intended potential with the support of appropriate guiding and policy frameworks. On this regard, progresses have been made in promoting responsible LSAI practices through international and regional guidelines, such as the Voluntary Guidelines on the Responsible Governance of Tenure (VGGT) (FAO (2019)) and Principles for Responsible Investment in Agriculture and Food Systems (RAIs) (CFS (2014)). Despite these progresses, the results of this study suggested that much remains to be done in the implementation of these guidelines to realize the transformational potential of large-scale land investments. For example, the Land Matrix's third analytical report found that VGGT implementation remains low, with only 25% of African deals meeting minimum standards and one-third failing to comply (Lay et al., 2021).

Impact of LSAIs varies by welfare indicators. The substantial heterogeneity observed in the intercept-only model (Panel A of Table 3) may arise from aggregating diverse welfare indicators together. To address this, we conducted a separate analysis based on specific welfare indicators which have sufficient number of effect size estimates in the meta-analysis dataset. Panel B of Table 3 presents these results. From these outcomes, LSAIs demonstrated statistically significant impacts on income (PCC = 0.051, 95% CI [−0.0039, 0.1054]) and food security (PCC = 0.026, 95% CI [−0.0091, 0.0616]), while the results for employment and assets are not significant. The moderate heterogeneity between effect sizes observed in the welfare-specific models suggests that welfare-specific analyses may better capture true effects than the pooled estimate.

To assess potential variation across different welfare outcomes, we examined differences in effect sizes among outcome types using Wald test. The Wald test results with HKSJ adjustment indicated no statistically significant variation across outcomes (F (4, 13) = 1.35, $p = 0.303$), which suggests LSAI impacts on community welfare remain stable regardless of the specific outcome measure used. The impact of LSAIs on individual estimates on the statistically significant welfare indicators such as income and food security of local communities, is presented in Figs. 3, 4 using random effect forest plots.

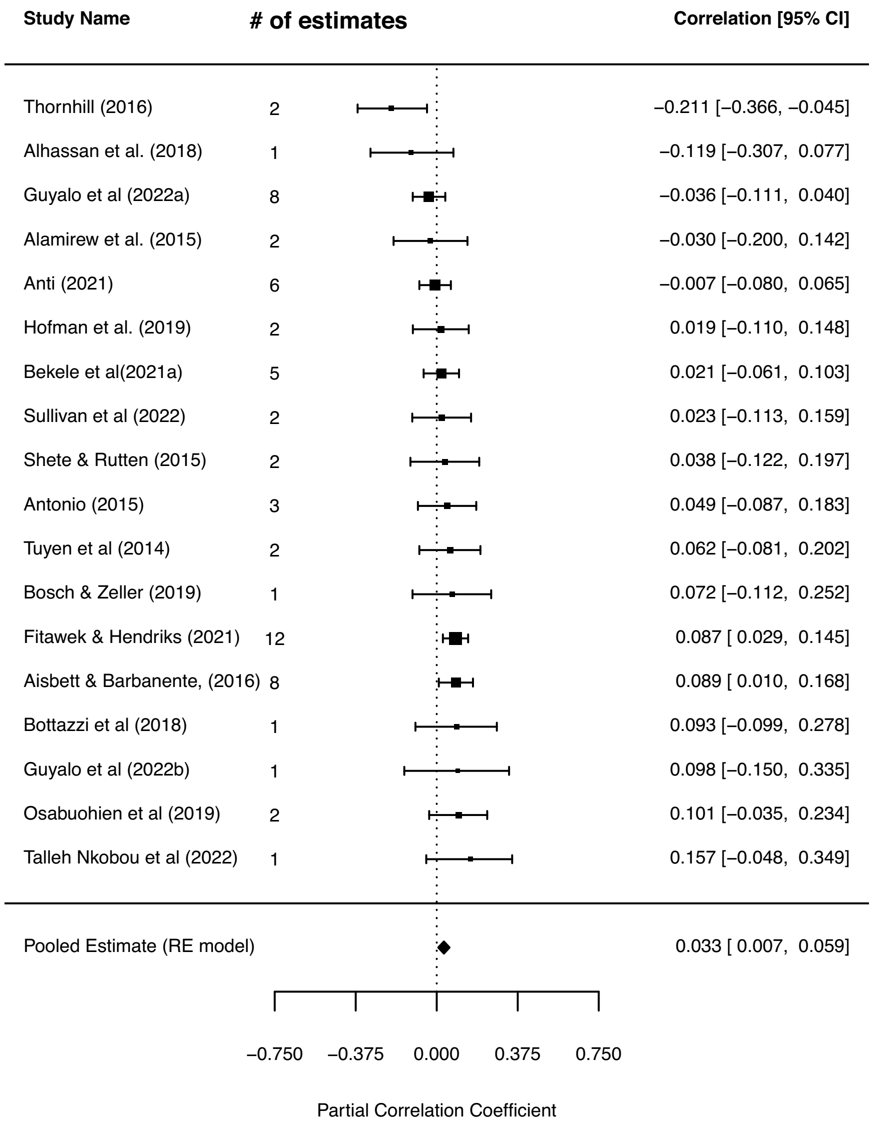

Fig. 4 Forest plot for random effects meta-analysis. It indicates the average effect of LSAIs on food security of local communities.

These results offer insights into the mechanisms through which LSAIs affect welfare of local communities. As indicated in the theoretical framework (Fig. 1), the impact of LSAIs on local community welfare depends on the extent of job creation, the availability of spillover effects, and availability of adequate compensation. The positive PCCs reported for income, food security, and employment outcomes, despite small, indicate that LSAIs could contribute for the local community welfare by improving income and food security through employment creation. However, negative impact of LSAIs on assets, despite statistically insignificant, may indicate adverse effects of LSAIs such as farm and grazing land dispossession. Empirical evidence suggests that LSAIs can negatively affect local communities through environmental degradation and resource loss (Jiao et al., 2015; Schoneveld, 2014), inadequate compensation, and social tensions (Cotula, 2009; Deininger & Xia, 2016). The primary implication of the results is that income and food security offer valid explanations for the observed overall positive effects.

The impact of LSAIs varies by production arrangements. Panel C of Table 3 presents results of the meta-analysis performed based on LSAI schemes. The results indicated that the impact of LSAIs on local community welfare depends on the production arrangements adopted. Different models, such as large-scale land acquisitions versus more integrated production systems like out-grower schemes and contract farming platforms, have varying effects on local communities. In our meta-analysis database, 23% of the studies and 22% of the estimates pertain to the impact of out-grower schemes on the welfare outcomes of local communities. Although studies focusing on out-grower schemes are fewer in number, the average PCC indicates that their impacts are substantially larger than those of plantation-based LSAIs. This suggests that despite the potential of LSAIs to promote rural development, practical experiences indicate significant limitations. One of the welfare gains to the local community materializes through the employment opportunities. However, these investments mostly rely on mechanized production systems, which require minimal labor and often displace local, labor-intensive smallholders. Additionally, the anticipated positive spillover effects have frequently failed to materialize at scale due to variations in local production systems (Ali et al., 2019; Lay et al., 2021).

Although out-grower-based LSAIs are not the only solution for the negative impacts of LSAIs (Lay et al., 2021), studies suggest that out-grower schemes, such as contract farming, provide benefits to local communities. These farming models help smallholder farmers to overcome previous constraints related to limited access to credit, improved inputs, and better output markets (Barrett et al., 2012). Improved access to these resources enhances productivity, and when combined with better market opportunities, it contributes to increased incomes for farm households. However, out-grower schemes are also associated with risks related to their arrangements which could reduce welfare. Smallholders participating in these schemes often face heightened exposure to production and marketing risks, such as those arising from the adoption of new crops or entry into unfamiliar markets (Eaton et al. (2001)).

Despite these risks, recent studies indicate that participants in out-grower-based LSAIs registered improved welfare and farm productivity outcomes. For example, Sullivan et al. (2022) evaluated the impact of contract farming opportunities from a large-scale agricultural investment on productivity in Tanzania. Their findings indicated that farm households engaged in contract farming achieved higher productivity due to access to improved inputs. Similarly, Herrmann (2017) studied the welfare effects of out-grower-based LSAIs in the Kilombero District of Tanzania. The results showed overall positive welfare differences between participants in the investments and their respective counterfactuals. In Malawi, Herrmann & Grote (2015) also found that participation in out-grower LSAIs led to improvements in household welfare, primarily through reductions in income poverty compared to the counterfactual.

The impact of LSAIs varies across countries hosting these investments. Our meta-analysis database comprises studies conducted across 15 countries: 13 are from Sub-Saharan Africa (SSA), and the remaining two from Asia. This reflects the concentration of LSAI flows to SSA, a region that has attracted significant number of investors due to the purported availability of arable land (Lay et al., 2021). Figure 5 portrays the random effect PCC estimates aggregated at country level using forest plots. The results reveal considerable variation in the welfare impacts of LSAIs across countries. The positive significant effect of LSAIs on the welfare of host communities holds for most host countries included in our synthesis. However, the estimated PCC for Cambodia, Mozambique, and Ethiopia turns out negative signaling the deteriorating welfare of host communities due to the advent of LSAIs.

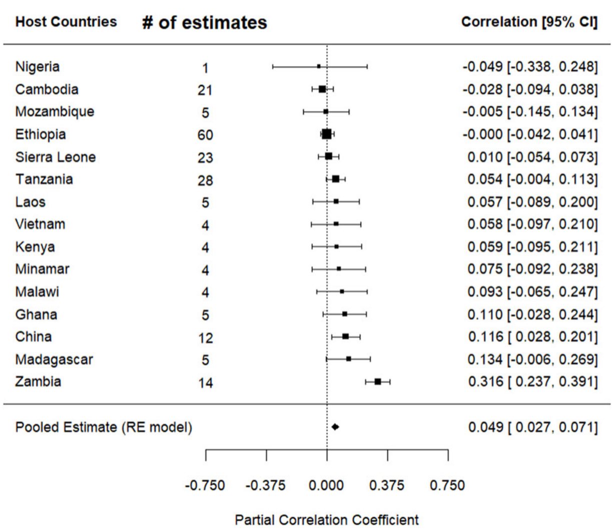

Fig. 5 Aggregated Forest plot. It summarizes the average impact of LSAIs on local communities by countries that host these investments.

The effect estimates of two countries, Ethiopia and Zambia, worth particular attention. Zambia registered an average PCC above the overall mean (0.316). This indicates that LSAIs have improved welfare in Zambia, suggesting that the country's policies and implementation strategies may offer valuable lessons for others. In contrast, the negative effect estimates for Ethiopia, where majority of the estimates are taken, indicate that LSAIs are deteriorating welfare of local communities. Given that Ethiopia is one of the most important target countries in SSA where more than 30 per cent of total land deals in Africa transpired (Lay et al., 2021), the negative impacts suggest that policies that promote large-scale agricultural investments should be discussed in-depth and weigh the pros and cons. The heterogeneity in the estimated PCC across countries could be attributed to the differences in the land tenure and land governance systems. Studies showed that LSAI mostly target countries with week institutional systems such as control of corruption (Bujko et al., 2016).

To evaluate the robustness of our findings, we conducted several sensitivity analyses, the results of which are presented in the supplementary material. First, we re-estimated the meta-analysis without winsorizing the data at the 5% level. The results (Table S2) indicated marginally higher effect sizes, suggesting that winsorization balances the distribution by mitigating the influence of extreme values. Second, we assessed the presence of influential observations using Cook's distance function (Cook (1979)). We then examined the sensitivity of our results by excluding these influential observations. The results (Table S3) showed minor deviations from the overall effect obtained in the intercept-only model, and were comparable to the results obtained when covariates were included in the winsorized dataset.

Publication bias test results

Funnel Plot. The funnel plot provides a visual inspection of publication bias, plotting the average PCC on the X-axis and precision (the inverse of the standard error of the PCC) on the Y-axis The funnel plot for the selected primary studies is shown in Fig. 6. In the absence of publication bias, studies are distributed symmetrically in an inverted funnel shape, indicating that the selection for publication is not systematically biased but instead reflects random sampling error (Borenstein et al., 2009). Even though the plots look like an inverted funnel, a closer look at them reveals that they are not symmetrical. It indicated that more

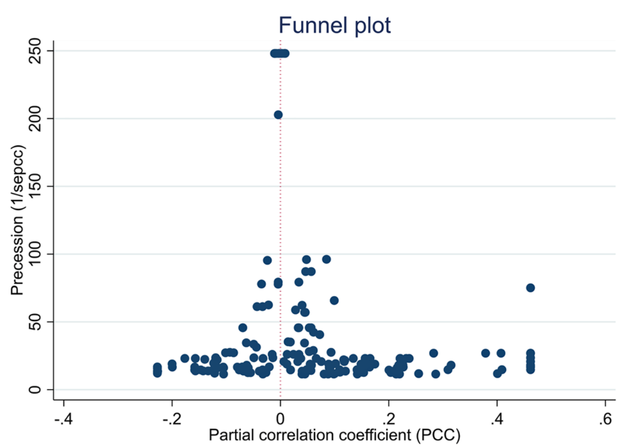

Fig. 6 Funnel plot of random effects meta-analysis. It examines publication bias on the literature that examine the impact of LSAIs on welfare of local communities.

Table 4 Linear and non-linear publication bias test results.

Panel A: FAT-PET	ME	FE	WLS
Publication bias (Constant)	2.729* (1.353)	0.245(0.980)	−3.422(2.203)
Mean beyond bias(1/SE)	0.0533**(0.0246)	0.0429**(0.0203)	0.620(0.332)
Observations	195	195	195
Number of Studies	39	39	39
Panel B: PEESE	ME	FE	WLS
Publication bias (SE)	−74.249(69.74)	−113.709(115.82)	−210.88*(111.19)
Mean beyond bias(1/SE)	0.0448**(0.0167)	0.0203*(0.012)	0.589**(0.289)
Observations	195	195	195
Number of Studies	39	39	39

Bootstrapped standard errors are presented in parenthesis.
ME mixed-effects multilevel model, *FE* fixed effect model, *WLS* weighted least squares.
*$p < 0.10$, **$p < 0.05$.

Table 5 Meta regression results explaining the heterogeneity of the average partial correlation coefficient.

Moderator variables	estimate	SE	z-value	95% CI
Intercept	0.0212	0.022	0.9647	[−0.0219, 0.0643]
Data characteristics				
Data type	0.0249	0.083	0.2996	[−0.1378, 0.1875]
Age of farm	0.0015	0.0075	0.1947	[−0.0132, 0.0161]
Publication characteristics				
Publication status	−0.2474***	0.0906	−2.7305	[−0.425, −0.0698]
citation count	0.0018*	0.001	1.8064	[−0.0001, 0.0037]
Relative publication year	0.0305*	0.0165	1.8494	[−0.0018, 0.0627]
Journal impact factor	0.0937*	0.0536	1.7482	[−0.0114, 0.1988]
Estimation methods				
Propensity score matching	−0.0292	0.0534	−0.5457	[−0.1339, 0.0756]
Difference-in-difference	−0.0291	0.0972	−0.2996	[−0.2196, 0.1614]
Ordinary least squares	−0.0777	0.0626	−1.2401	[−0.2005, 0.0451]
Outcomes				
Asset	−0.0485	0.0423	−1.145	[−0.1314, 0.0345]
Employment	0.0937*	0.052	1.8009	[−0.0083, 0.1957]
Food security	0.0614	0.0386	1.5936	[−0.0141, 0.137]
Income	0.0447	0.0381	1.1753	[−0.0299, 0.1193]
Livelihoods	0.072	0.07	1.0299	[−0.0651, 0.2092]
Countries				
Cambodia	−0.1045	0.0821	−1.2726	[−0.2655, 0.0565]
China	0.0557	0.0738	0.755	[−0.0889, 0.2002]
Ethiopia	−0.0764	0.0502	−1.5218	[−0.1747, 0.022]
Sierra Leone	−0.1047	0.0828	−1.2641	[−0.2671, 0.0576]
Tanzania	−0.1113*	0.065	−1.7131	[−0.2386, 0.016]
Zambia	0.3585***	0.1005	3.5683	[0.1616, 0.5554]
Author affiliation	−0.02	0.0541	−0.3692	[−0.1261, 0.0861]
LSAI schemes				
Out grower scheme	0.0424	0.0628	0.6756	[−0.0806, 0.1655]
Model statistics				
Between study variance (τ^2)		0.0015 (SD = 0.0383)		
Within study variance (σ^2)		0.0150 (SD = 0.1224)		
Test of Moderators (QM Test)		QM (df = 22) = 58.196, p < .0001		

The response variable in this analysis is the partial correlation coefficient (PCC). The meta regression model is estimated using robust variance estimation meta-analysis using the correlated and hierarchical (CHE) working model. The intercept indicates the overall effect after the covariates are accounted. For a detailed description of all the variables included in the analysis, please refer to Table 2.
SE robust standard error, *SD* standard deviations, *CI* confidence interval, *QM* Cochran's Q statistic for moderators.
***$p < 0.01$, and *$p < 0.10$.

positive estimates could probably be preferentially reported. Visual inspection of the funnel plot, however, do not provide the extent of publication bias and thus conclusive evidence for the presence of publication bias.

Table 4 presents results from the linear and non-linear publication bias tests. Panel A of Table 4 presents the linear publication bias test results based on Eq. (6). The first column captures results from the mixed-effect model. The publication bias coefficient is not significant at 5% level, indicating a absences of publication bias. The publication bias corrected average PCC (0.0533) is also comparable to the magnitude obtained from the RVE model (see Table 3). The results of the fixed effect model are presented in column two. It suggests no publication bias, with a mean effect beyond publication bias of 0.0429. While the results of the mean effect beyond publication bias are robust across the mixed-effect and fixed effect models, the weighted least squares model in the third column are not robust.

Panel B in Table 4 presents the non-linear publication bias test results based on Eq. (7). While the results of the mixed-effect (ME) and fixed-effect (FE) models indicate no evidence of publication bias, the publication bias-corrected mean PCC in the fixed-effects model is less precise. Similar to the linear model, the WLS model in the non-linear specification is not robust, suggesting an exaggerated publication bias-corrected mean PCC.

In general, while our analysis was not specifically aimed at extensively evaluating publication bias in the LSAI literature, the results from both linear and non-linear methods suggest an absence of publication bias in the primary studies included in our meta-analysis. Furthermore, the publication bias corrected-mean PCC is positive ranging from 0.002 to 0.0533 suggesting a favorable effect of LSAIs on the welfare of local communities.

Meta-regression results. The box plot in Fig. 2 and the overall effect size results in Table 3 demonstrated significant heterogeneity. To understand the potential sources of the heterogeneity, we conducted a meta-regression analysis (MRA) using Eq. (3), estimated via the robust variance estimation (RVE) with the correlated and hierarchical effects (CHE) working model. Description and summary statistics of the moderator variables used in the MRA are presented in Table 2. To mitigate multicollinearity problem and enhance interpretability, the moderator variables are centered around their means.

Table 5 presents the MRA results. The model statistics indicated a very small between study heterogeneity ($\tau^2 = 0.0015$, $SD = 0.0383$) and moderate within-study heterogeneity ($\sigma^2 = 0.0150$, $SD = 0.1224$), suggesting that within study variability contributed more to heterogeneity than the between study variability. The significant QM joint test of moderators indicates that the moderators included explains a substantial portion of heterogeneity of the effect size.

Among the categories of moderators included, publication status significantly explains the heterogeneity in the effect size. Specifically, published studies reported relatively smaller effect sizes compared to unpublished studies. More highly cited studies showed slightly larger effect sizes, and studies published in journals with impact factors exhibited a positive impact on effect size. This suggests that the peer-review process affects the magnitude of effect sizes reported in primary studies. Additionally, relative publication year, proxied by the difference between publication year and the earliest publication year in the dataset, positively influenced effect sizes, indicating that newer studies may report slightly larger effect estimates than older studies. Studies evaluating the impact of LSAIs on employment reported relatively larger effect estimates. Regarding study locations, estimates from studies focusing on LSAIs in Zambia explained a significant portion of the heterogeneity, as also evidenced by the host country subgroup analysis (Fig. 5), which showed that studies in Zambia reported an average effect size exceeding the overall mean. Methodological variables (data type, estimation methods) and LSAI schemes did not significantly affect effect sizes.

Conclusion

In recent years, governments across developing countries have promoted large-scale agricultural investments (LSAIs) with an aim of revitalizing the agricultural sector and enhance economic development. Proponents argue that these investments can stimulate economic development, generate employment, and boost agricultural productivity, contributing to overall welfare improvements in host communities. However, empirical studies have reported both positive and negative consequences of these investments which make it challenging to draw generalizable conclusions about the broader effects of LSAIs. To address this, we conducted a meta-analysis, synthesizing 195 effect size estimates extracted from 39 studies to provide a comprehensive assessment of the overall effects of LSAIs on the welfare of local communities.

Our synthesis revealed a statistically positive but economically negligible average partial correlation coefficient (PCC) of 0.043. This minimal effect size, explaining only less than 0.2% of welfare variation ($(PCC)^2 = 0.0018$), suggests that LSAIs have limited practical impact on improving living standards in host communities. Furthermore, we observed significant heterogeneity in LSAI impacts across host countries, indicating that a uniform approach to designing and implementing government programs promoting LSAIs may be counterproductive. Primary studies do not provide information why the impact of LSAIs is heterogenous across countries. Evidence generated by the primary studies focused from assessment of a single investment project which makes it difficult to generalize findings beyond the community level where LSAIs carried out. Future studies could focus cross country study to explain this heterogeneity.

The observed small mean effect size and cross-country heterogeneity may be attributed to poor implementation, monitoring, and evaluation of LSAIs. However, the positive sign suggests that LSAIs could achieve their intended potential if governments follow up the appropriate implementation of guiding frameworks and policies, such as the Voluntary Guidelines on the Responsible Governance of Tenure (VGGTs). Additionally, our analysis suggests that out-grower schemes and contract farming platforms may offer greater benefits than plantation-based land acquisitions. Therefore, governments in developing countries should prioritize these more integrated models. Moreover, most of the studies included in our synthesis evaluated the short-term effects of LSAIs using cross sectional data which limit our understanding on the long-term impacts of LSAIs. However, the impact of such investments would turn out differently in the long term. For instance, Hofman et al. (2019) found that LSAIs led to a more significant reduction in income over the long term than in the short term. Furthermore, we observed that the overall impact of LSAIs at a national level, that could accrue in the form of increased agricultural production, agricultural exports, tax revenue, etc., didn't get much attention in the existing literature. Future studies could also focus in accounting the impact of LSAIs at national level. Finally, the primary studies included in the synthesis have focused on evaluating operational land deals. However, non-operational deals also have significant potential and actual effect on local communities (Borras et al., 2022). Future studies could explore in studying the impact of non-operational large scale agricultural investments on local communities.

The limitation of this meta-analysis is that we did not conduct a formal critical appraisal of the included studies. While we performed publication bias tests, we could not account for the possibility of low-quality studies having an undue influence on the results of this meta-analysis. Therefore, we are unable to fully assess the risk of bias or the methodological quality of the primary studies. This may limit our overall findings and suggests that the results should be interpreted with caution.

Data availability

The data supporting this study's findings are available from the corresponding author upon reasonable request.

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Notes

1 Land matrix is the most comprehensive and the most used database on land deals (available at www.landmatrix.org). However, land deals reported in the land Matrix data set are biased downwards due to under reporting of deals by host countries. Thus, the figures indicate the extreme lower bound (Lay et al., 2021).

2 After we conducted the search and coding processes, we became aware about an updated meta-analysis guideline (Irsova et al., 2023). We try to ensure that the remaining procedures are in line with the updated guideline.

3 There are two alternative approaches (Stanley & Doucouliagos, 2012). The first is taking the "best-set" estimates, i.e., extracting estimates from regression results preferred by the author. However, in many instances, authors may not reveal their preferred estimates, and even if they do, taking only the preferred estimates may result in publication selection bias. Thus, the best-set estimates might not be the best data set to be used in meta-analysis. The second is taking the "average-set" estimates, i.e., taking the average of all the reported estimates of each study. However, this approach conceals the within study variation-a relevant information in meta-analysis.

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LSA: Conceptualization, Methodology, Software, Validation, Data curation, Formal Analysis, writing—Original Draft, Visualization; WHS: Supervision, Writing-Review and Editing, Funding acquisition; SZW: Conceptualization, Methodology, Writing-Review and Editing, Visualization; WAK: Conceptualization, Data Curation, Visualization, Writing-Review and Editing; FB: Methodology and Data Curation. All authors listed have made direct contribution to the work and approved it for publication.

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